# Лабораторная работа №5

Расширенная настройка HTTP-сервера Apache

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***дисциплина:*** **Администрирование сетевых подсистем**

# Цель работы

Приобретение практических навыков по расширенному конфигурированию HTTP-сервера Apache в части безопасности и возможности использования PHP.

root@server:/etc/pki/tls/private

```
[vagrant@server ~]$ sudo -i
[root@server ~]# mkdir -p /etc/pki/tls/private
[root@server ~]# ln -s /etc/pki/tls/private /etc/ssl/private
[root@server ~]# cd /etc/pki/tls/private
[root@server private]# openssl req -x509 -nodes -newkey rsa:2048 -keyout www.use
r.net.key -out www.user.net.crt
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```

root@server:/etc/pki/tls/private

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-----
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:
State or Province Name (full name) []:
Locality Name (eg, city) [Default City]:
Organization Name (eg, company) [Default Company Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (eg, your name or your server's hostname) []:
Email Address []:
[root@server private]#
[root@server private]#
```

root@server:/etc/pki/tls/private

```
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You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:RU
State or Province Name (full name) []:Russia
Locality Name (eg, city) [Default City]:Moscow
Organization Name (eg, company) [Default Company Ltd]:vagrant
Organizational Unit Name (eg, section) []:vagrant
Common Name (eg, your name or your server's hostname) []:user.net
Email Address []:user@user.net
[root@server private]#
```

Open Save

**www.user.net.conf**
/etc/httpd/conf.d

```
<VirtualHost *:80>
ServerAdmin webmaster@user.net
DocumentRoot /var/www/html/www.user.net
ServerName www.user.net
ServerAlias www.user.net
ErrorLog logs/www.user.net-error_log
CustomLog logs/www.user.net-access_log common
RewriteEngine on
RewriteRule ^(.*)$ https://%{HTTP_HOST}$1 [R=301,L]
</VirtualHost>
<IfModule mod_ssl.c>
<VirtualHost *:443>
SSLEngine on
ServerAdmin webmaster@user.net
DocumentRoot /var/www/html/www.user.net
ServerName www.user.net
ServerAlias www.user.net
ErrorLog logs/www.user.net-error_log
CustomLog logs/www.user.net-access_log common
SSLCertificateFile /etc/ssl/certs/www.user.net.crt
SSLCertificateKeyFile /etc/ssl/private/www.user.net.key
</VirtualHost>
</IfModule>
```

root@server:/etc/httpd/conf.d

```
 Failed to execute child process “dbus-launch” (No such file or directory)
[root@server conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http imap imaps pop3 pop3s smtp ssh
[root@server conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp am
qps apcupsd audit ausweisapp2 bacula bacula-client bareos-director bareos-fileda
emon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-r
pc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit col
lectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dh
cpv6-client distcc dns dns-over-tls docker-registry docker-swarm dropbox-lansync
 elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 fr
eeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-cl
ient ganglia-master git gpsd grafana gre high-availability http http3 https iden
t imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin
 kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver
kube-control-plane kube-control-plane-secure kube-controller-manager kube-contro
ller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure
kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-t
ls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix m
dns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysq
l nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvp
n ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi
pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis re
```

root@server:/etc/httpd/conf.d

```
dns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysq
l nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvp
n ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi
pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis re
dis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client sam
ba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snm
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn sy
ncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle
tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server warpi
nator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discove
ry-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp
-server zabbix-agent zabbix-server zerotier
[root@server conf.d]# firewall-cmd --add-service=https
success
[root@server conf.d]# firewall-cmd --add-service=https --permanent
success
[root@server conf.d]# firewall-cmd --reload
```

```
[root@server conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp am
qps apcupsd audit ausweisapp2 bacula bacula-client bareos-director bareos-fileda
emon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-r
pc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit col
lectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dh
cpv6-client distcc dns dns-over-tls docker-registry docker-swarm dropbox-lansync
 elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 fr
eeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-cl
  nt ganglia-master git gpsd grafana gre high-availability http http3 https ider
  imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmir
 kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver
kube-control-plane kube-control-plane-secure kube-controller-manager kube-contro
ller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure
kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-t
ls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix m
dns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysq
l nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvp
n ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi
pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
```

Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share

# Hmm. We're having trouble finding that site.

We can't connect to the server at www.user.net.

**If you entered the right address, you can:**

- Try again later

root@server:/etc/httpd/conf.d

```
[root@server conf.d]# systemctl restart httpd
Job for httpd.service failed because the control process exited with error code.
See "systemctl status httpd.service" and "journalctl -xeu httpd.service" for details.
[root@server conf.d]# dnf -y install php
Last metadata expiration check: 0:56:54 ago on Thu 05 Mar 2026 06:39:58 PM UTC.
Dependencies resolved.
================================================================================
 Package                Arch        Version                  Repository      Size
================================================================================
Installing:
 php                    x86_64      8.0.30-5.el9_7           appstream      8.8 k
Installing dependencies:
 nginx-filesystem       noarch      2:1.20.1-24.el9          appstream      9.2 k
 php-common             x86_64      8.0.30-5.el9_7           appstream      667 k
Installing weak dependencies:
 php-cli                x86_64      8.0.30-5.el9_7           appstream      3.1 M
 php-fpm                x86_64      8.0.30-5.el9_7           appstream      1.6 M
 php-mbstring           x86_64      8.0.30-5.el9_7           appstream      468 k
 php-opcache            x86_64      8.0.30-5.el9_7           appstream      510 k
 php-pdo                x86_64      8.0.30-5.el9_7           appstream       82 k
 php-xml                x86_64      8.0.30-5.el9_7           appstream      133 k

Transaction Summary
================================================================================
Install  9 Packages

Total download size: 6.5 M
Installed size: 35 M
Downloading Packages:
(1/9): nginx-filesystem-1.20.1-24.el9.noarch.rp  29 kB/s | 9.2 kB     00:00
(2/9): php-8.0.30-5.el9_7.x86_64.rpm             19 kB/s | 8.8 kB     00:00
(3/9): php-cli-8.0.30-5.el9_7.x86_64.rpm        3.1 MB/s | 3.1 MB     00:00
(4/9): php-common-8.0.30-5.el9_7.x86_64.rpm     536 kB/s | 667 kB     00:01
(5/9): php-mbstring-8.0.30-5.el9_7.x86_64.rpm   536 kB/s | 468 kB     00:00
(6/9): php-opcache-8.0.30-5.el9_7.x86_64.rpm    860 kB/s | 510 kB     00:00
(7/9): php-fpm-8.0.30-5.el9_7.x86_64.rpm        931 kB/s | 1.6 MB     00:01
(8/9): php-pdo-8.0.30-5.el9_7.x86_64.rpm        245 kB/s |  82 kB     00:00
(9/9): php-xml-8.0.30-5.el9_7.x86_64.rpm        793 kB/s | 133 kB     00:00
```

```
** (gedit:9634): WARNING **: 19:39:11.957: Set document metadata failed: Setting attribute metadata::gedit-en
coding not supported

** (gedit:9634): WARNING **: 19:39:12.027: Set document metadata failed: Setting attribute metadata::gedit-sp
ell-language not supported

** (gedit:9634): WARNING **: 19:39:12.027: Set document metadata failed: Setting attribute metadata::gedit-en
coding not supported

** (gedit:9634): WARNING **: 19:39:25.012: Set document metadata failed: Setting attribute metadata::gedit-po
sition not supported

(gedit:9634): dconf-WARNING **: 19:39:25.037: failed to commit changes to dconf: Failed to execute child proc
ess “dbus-launch” (No such file or directory)
[root@server www.user.net]# chown -R apache:apache /var/www
[root@server www.user.net]# restorecon -vR /etc
[root@server www.user.net]# restorecon -vR /var/www
[root@server www.user.net]# systemctl restart httpd
Job for httpd.service failed because the control process exited with error code.
See "systemctl status httpd.service" and "journalctl -xeu httpd.service" for details.
[root@server www.user.net]#
```

root@server:/var/www/html/www.user.net

```
** (gedit:9634): WARNING **: 19:39:12.027: Set document metadata failed: Setting attribute metadata::gedit-sp
ell-language not supported

** (gedit:9634): WARNING **: 19:39:12.027: Set document metadata failed: Setting attribute metadata::gedit-en
coding not supported

** (gedit:9634): WARNING **: 19:39:25.012: Set document metadata failed: Setting attribute metadata::gedit-po
sition not supported

(gedit:9634): dconf-WARNING **: 19:39:25.037: failed to commit changes to dconf: Failed to execute child proc
ess “dbus-launch” (No such file or directory)
[root@server www.user.net]# chown -R apache:apache /var/www
[root@server www.user.net]# restorecon -vR /etc
[root@server www.user.net]# restorecon -vR /var/www
[root@server www.user.net]# systemctl restart httpd
Job for httpd.service failed because the control process exited with error code.
See "systemctl status httpd.service" and "journalctl -xeu httpd.service" for details.
[root@server www.user.net]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/autoindex.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/fcgid.conf'? Y
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/manual.conf'? yes
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/README'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/rocky-snipolicy.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/server.user.net.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/ssl.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/user.net'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/welcome.conf'?
cp: overwrite '/vagrant/provision/server/http/etc/httpd/conf.d/www.user.net.conf'?
[root@server www.user.net]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
[root@server www.user.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/private
[root@server www.user.net]# mkdir -p /vagrant/provision/server/http/etc/pki/tls/certs
[root@server www.user.net]# cp -R /etc/pki/tls/private/www.user.net.key /vagrant/provision/server/http/etc/pk
i/tls/private
[root@server www.user.net]# cp -R /etc/pki/tls/certs/www.user.net.crt /vagrant/provision/server/http/etc/pki/
tls/certs
[root@server www.user.net]#
```

# *Вывод*

В ходе выполнения лабораторной работы были приобретены практические навыки по расширенному конфигурированию HTTP-сервера Apache в части безопасности и возможности использования PHP.

*Спасибо за внимание!*